

MADHAV MERUVA

Backend Engineer | Distributed Systems & Scalable APIs

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SUMMARY

Computer Science undergraduate specializing in **backend engineering and scalable application** development, with hands-on experience building **REST APIs**, distributed **backend services**, and real-time systems. Improved API performance by **~30%** through **Redis caching, pagination, and optimization** techniques. Experienced in designing **modular** architectures, database-driven applications, and production-ready backend systems using Java, Python, Node.js, and FastAPI. Seeking Backend / Software Engineering internship opportunities.

EDUCATION

Keshav Memorial Institute of Technology (KMIT)

Bachelor of Technology, Computer Science Engineering | 2023 - 2027 | CGPA - 8.5

Sri Chaitanya Junior College

Intermediate (MPC) | 2021-2023 | GPA - 9.8

PROJECTS

Startup Idea Validator (Distributed Backend System) | Aug 2025 - Oct 2025

Live: startupideavalidator.vercel.app | **GitHub:** github.com/mad674/startupideavalidator

- Designed and deployed **scalable backend services** for a real-time AI evaluation platform **supporting semantic retrieval** and optimized API workflows.
- Improved API response time by **~30% using Redis caching, idempotent APIs, and pagination** strategies.
- Built modular REST APIs integrating FastAPI, Express.js, MongoDB, and vector search systems.
- Implemented **JWT-based authentication** and scalable service architecture for concurrent user interactions.

Technologies: React.js, Node.js (Express), FastAPI, MongoDB, PineconeDB, Redis, LangChain, OpenAI API, JWT

Financial Question Answering System (FinQA) | Feb 2025 - Jun 2025

Live: financeqptapp.netlify.app | **GitHub:** github.com/mad674/qa

- Engineered an end-to-end **backend inference pipeline** for a financial question-answering system using FastAPI and scalable inference workflows.
- Developed modular **preprocessing, training, and inference** workflows for scalable ML integration.
- Achieved **~80% accuracy** using interpretable program-step reasoning and symbolic execution techniques.

Technologies: Flutter, FastAPI, PyTorch, Firebase

Jewellery Design Pattern Generator (Pix2Pix GAN) | Aug 2024 - Dec 2024

Live: jewelrydesignpatternngen.netlify.app | **GitHub:** github.com/mad674/mlmodel

- Built a **Pix2Pix GAN-based system for sketch-to-image generation** with real-time inference capabilities
- Developed a **real-time pipeline for generating design previews** from user-uploaded sketches
- Improved output quality through **dataset preprocessing** and training optimization techniques

Technologies: React.js, Node.js (Express), JWT, MongoDB, TensorFlow, Flask

SKILLS

- Languages:** Java, Python, C++, JavaScript
- Backend:** Node.js (Express), FastAPI, REST APIs, JWT Authentication
- Databases:** MongoDB, MySQL, Vector Databases, Redis
- AI/ML:** PyTorch, TensorFlow, NLP, LangChain
- Cloud & DevOps:** Git, Postman, AWS, Docker
- Core Concepts:** DSA, OOP, OS, DBMS, Computer Networks

ACHIEVEMENTS AND CERTIFICATES

- Solved **500+ DSA** problems across arrays, graphs, trees, and dynamic programming
- Secured a **top team award** at the KMIT Project Expo